Abiral Shrestha

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Education

Gettysburg College — Gettysburg, PA
Bachelor of Science in Computer Science, Minor in Mathematics — *Aug 2024 – May 2028*

Experience

Software Developer — May 2025 – Present
Association for Computing Machinery, Gettysburg College — *Gettysburg, PA*

- Engineered the frontend for "TexQuest," a LeetCode-style competitive LaTeX platform used by the CS department, utilizing React and Framer Motion to support real-time rendering and improve load performance by 30%
- Developed a secure Admin Dashboard to manage live contests, users, and submissions, implementing RBAC (Role-Based Access Control) that reduced manual administrative overhead by 40%.
- Redesigned core pages for clarity and accessibility, increasing student engagement and platform usage by 25%.

Tech Team — Aug 2025 – Present
Office of Student Activities and Greek Life, Gettysburg College — *Gettysburg, PA*

- Set up and operated sound, lighting, and projection systems to support campus events with 200+ attendees.
- Minimized event disruptions by troubleshooting and resolving equipment issues under pressure, ensuring 100% continuity.

Projects

Foot-Steps | *React, PostgreSQL, AWS S3, Docker, REST APIs* — Jan 2026 – Feb 2026

- Developed full-stack travel app with React frontend and Node.js/Express backend, automating photo organization into timeline and map views for 100+ photo trips in under 2 minutes.
- Designed cloud-based media pipeline with AWS S3 storage, Prisma ORM, and automatic EXIF metadata extraction to generate GPS-based map pins without manual tagging.
- Containerized frontend and backend services using Docker to ensure 100% environment consistency across development and production.

Inhibitor | *TypeScript, React* — Feb 2026

- Developed a Manifest V3 extension in React and TypeScript, coordinating popup, background, and content scripts via Chrome APIs and handling YouTube's dynamic DOM.
- Implemented intent-based focus sessions with automatic end-time and duration logging, removing the need for manual time tracking.

Graph Analysis Automation | *Flask, TypeScript, Docker, MySQL* — Nov 2025
Schneider Prize for Technology Innovation 2025

- Engineered an automated ETL pipeline for MRG Labs, processing 100+ spectral analysis CSV files instantly and reducing manual workflow time by 70%.
- Developed statistical algorithms to detect oxidation trends with 90% validated accuracy, replacing legacy Excel-based processes.
- Built an interactive dashboard using TypeScript that overlays spectral graphs for real-time comparison, enabling engineers to visually pinpoint data anomalies instantly.

Hennest | *React, Clerk, Supabase, TailwindCSS* — Mar 2025 - Apr 2025

- Built Hennest, a financial management app using React, Supabase, and Tailwind that lets users connect bank accounts and track spending across 1,000+ transactions.
- Integrated financial APIs and Gemini AI to categorize expenses into 12+ spending groups and generate personalized monthly summaries and saving insights.

Technical Skills

Languages: Java, Python, C, C#, SQL, JavaScript, TypeScript
Frameworks: React, Redux Toolkit, Express.js, JUnit, Flask, Pandas, NumPy, Matplotlib
Developer Tools: AWS S3, Git, Postman, Docker, Supabase, Clerk
Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Software Engineering, Discrete Mathematics, Linear Algebra